Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim:-Simulate dilation, erosion, opening and closing operation on images.**	
Experiment:-11	**Date:31-10-2026**	**Enrollment No:92400133063**

Code:-

```
import cv2
import numpy as np
import os
from google.colab.patches import cv2_imshow
# Define the image path (fallback to the existing images.jpeg if the certificate isn't
there)
image_path = '/content/images.jpeg'
if not os.path.exists(image_path):
    image_path = 'images.jpeg'
# Load the image
image = cv2.imread(image_path, 0)  # Load as grayscale
if image is None:
    print(f"Error: Could not load the image from '{image_path}'. Please check if
the file exists.")
else:
    # Define the kernel (structuring element)
    kernel = np.ones((5, 5), np.uint8)
    # Perform Erosion
    erosion = cv2.erode(image, kernel, iterations=1)
    # Perform Dilation
    dilation = cv2.dilate(image, kernel, iterations=1)
    # Perform Opening (Erosion followed by Dilation)
    opening = cv2.morphologyEx(image, cv2.MORPH_OPEN, kernel)
    # Perform Closing (Dilation followed by Erosion)
    closing = cv2.morphologyEx(image, cv2.MORPH_CLOSE, kernel)
    # Display the original image and the results
    print("Original Image:")
    cv2_imshow(image)
    print("Erosion:")
    cv2_imshow(erosion)
```

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```
print("Dilation:")
cv2_imshow(dilation)
print("Opening:")
cv2_imshow(opening)
print("Closing:")
cv2_imshow(closing)
```

output:-



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Conclusion:-

The given experiment fundamental morphological operations—erosion, dilation, opening, and closing—using a 5×5 structuring element on a grayscale image. Erosion shrinks bright regions by removing boundary pixels, while dilation expands them, making these operations useful for noise removal and filling gaps respectively. Opening (erosion followed by dilation) effectively removes small noise/objects while preserving larger structures, and closing (dilation followed by erosion) fills small holes and connects nearby regions. Together, these operations illustrate how kernel-based morphological transformations can be used for shape analysis, noise reduction, and structural refinement in image preprocessing tasks.